

6 Time (s)

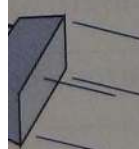
ough a device varies  
rough the device at

en by  $i(t) = 0.05t$  A.  
the device between

device is  $i(t) = 10$  A.  
 $i(t) = 20 - 2t$  A, and  
me and find the total  
0 s and  $t = 10$  s.

ice is  $q(t) = 1 - e^{-500t}$   
device versus time for

Figure P1-13 has an  
rs (Ah) when con-  
200 watts of power.  
it.  
battery.  
the headlight?



- 1-17 When illuminated, the  $i-v$  relationship for a photo-cell is  $i = e^v - 10$  A. For  $v = -2, 2$ , and  $3$  V, find the device power and state whether it is absorbing or delivering power.
- 1-18 A new 1.5-V AA battery delivers 40 kJ of energy during its lifetime. How long will the battery last in an application that draws 10 mA continuously. Assume the battery voltage is constant.
- 1-19 The maximum power the device can dissipate is 0.5 W. Determine the maximum current allowed by the device power rating when the voltage is 15 V.
- 1-20 A constant current of 2 A charges a battery for 4 hours. During the charging period, the battery voltage is  $v(t) = 12 - 2e^{-t}$ , where  $t$  is in hours. Determine the energy stored in the battery.
- 1-21 Two electrical devices are connected as shown in Figure P1-21. Using the reference marks shown in the figure, find the power transferred and state whether the power is transferred from A to B or B to A when
- (a)  $v = +33$  V and  $i = -2.2$  A
  - (b)  $v = -12$  V and  $i = -1.2$  mA
  - (c)  $v = +37.5$  V and  $i = +40$  mA
  - (d)  $v = -15$  V and  $i = -43$  mA

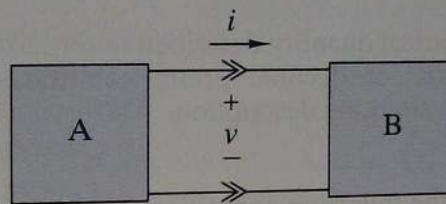


FIGURE P1-21

- 1-22 Figure P1-22 shows an electric circuit with a voltage and a current variable assigned to each of the six devices. The device voltages and currents are observed to be

|          | $v$ (V) | $i$ (A) |
|----------|---------|---------|
| Device 1 | 15      | -1      |
| Device 2 | 5       | 1       |
| Device 3 | 10      | 2       |
| Device 4 | -10     | -1      |
| Device 5 | 20      | -3      |
| Device 6 | 20      | 2       |

### INTEGRATING PROBLEMS

1-23 Power Ratio in dB

In complete darkness the voltage across and current through a two-terminal light detector are +5.6 V and +8 nA. In full



## PROBLEMS

### OBJECTIVE 2-1 ELEMENT CONSTRAINTS (SECT. 2-1)

Given a two-terminal element with one or more electrical variables specified, use the element  $i$ - $v$  constraint to find the magnitude and direction of the unknown variables. See Example 2-1

2-1 The current through a  $5\text{-k}\Omega$  resistor is  $12\text{ mA}$ . Find the voltage across the resistor.

2-2 A  $6.2\text{-k}\Omega$  resistor dissipates  $12\text{ mW}$ . Find the current through the resistor.

2-3 The voltage across and current through a resistor are  $10\text{ V}$  and  $2.5\text{ mA}$ . Find the conductance of the resistor.

2-4 In Figure P2-4 the resistor dissipates  $25\text{ mW}$ . Find  $R_x$ .

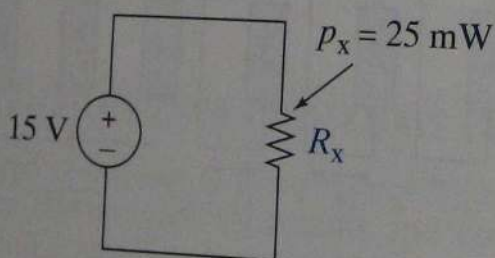


FIGURE P2-4

2-5 In Figure P2-5 find  $R_x$  and the power delivered to the resistor.

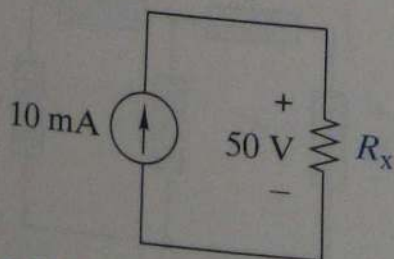


FIGURE P2-5

2-6 The  $i$ - $v$  relationship  $v = 75i + 0.2i^3$ .

(a) Calculate  $v$  and  $p$  for  $i = \pm 10\text{ A}$ .

(b) Find the maximum energy stored in the element treated as a  $75\text{-}\Omega$  linear resistor.  $|i| < 0.5\text{ A}$ .

2-7 A  $10\text{-k}\Omega$  resistor has a power rating of  $1\text{ W}$ . Find the maximum voltage that can be applied across the resistor.

2-8 A certain type of film resistor has resistance values between  $10\text{ }\Omega$  and  $10\text{ M}\Omega$ . Show that the voltage rating  $V$  is proportional to  $\sqrt{R}$  for  $R > 1\text{ M}\Omega$ , and that the power rating  $P$  is proportional to  $\sqrt{R}$  for  $R < 1\text{ M}\Omega$ .

2-9 Figure P2-9 shows the circuit model for a specific pn junction diode is

$$i = 2 \times 10^{-10} e^{v/0.025} - 2 \times 10^{-10} e^{-v/0.025}$$

- Explanations:  
 $P = i \cdot V$   
 $\frac{dP}{dV} = \frac{di}{dV} \cdot V + i$   
 $= V \cdot i$
- (a) Use this equation to find the power dissipated in the diode for  $v = \pm 0.2, \pm 0.4, \text{ and } \pm 0.8\text{ V}$ . Use the characteristic of the element to determine if the element is linear or nonlinear, and active or passive.
- (b) Is the diode linear or nonlinear, and active or passive?
- (c) Use the diode model to find the power dissipated in the diode. Do you think the model is accurate? Explain.
- (d) Repeat part (c) for  $v = -5\text{ V}$ .

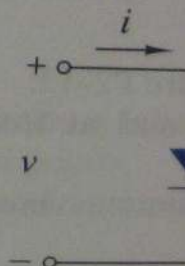


FIGURE P2-9

## OBJECTIVE 2-2 CONNECTION CONSTRAINTS (SECT. 2-2)

Given a circuit composed of two-terminal elements:

- Identify nodes and loops in the circuit.
- Identify elements connected in series and in parallel.
- Use Kirchhoff's laws (KCL and KVL) to find selected signal variables.

See Examples 2-4, 2-5, 2-6 and Exercises 2-1, 2-2, 2-3, 2-4, 2-5

2-10 In Figure P2-10  $i_2 = 2$  A and  $i_3 = -5$  A. Find  $i_1$  and  $i_4$ .

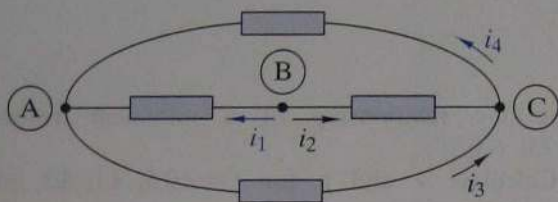


FIGURE P2-10

2-11 For the circuit in Figure P2-11:

- Identify the nodes and at least two loops.
- Identify any elements connected in series or in parallel.
- Write KCL and KVL connection equations for the circuit.

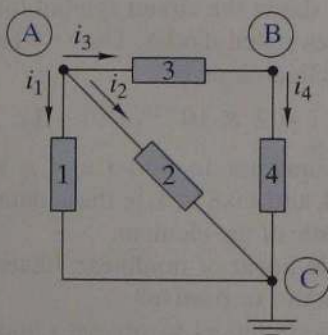


FIGURE P2-11

2-12 In Figure P2-11  $i_2 = 10$  mA and  $i_4 = 20$  mA. Find  $i_1$  and  $i_3$ .

2-13 For the circuit in Figure P2-13:

- Identify the nodes and at least three loops in the circuit.
- Identify any elements connected in series or in parallel.
- Write KCL and KVL connection equations for the circuit.

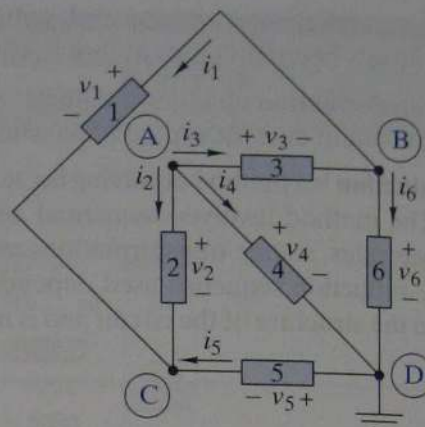


FIGURE P2-13

2-14 In Figure P2-13  $v_2 = 5$  V,  $v_3 = -8$  V, and  $v_4 = 3$  V. Find  $v_1$ ,  $v_5$ , and  $v_6$ .

2-15 The circuit in Figure P2-15 is organized around the three signal lines A, B, and C.

- Identify the nodes and at least three loops in the circuit.
- Write KCL connection equations for the circuit.
- If  $i_1 = -20$  mA,  $i_2 = -12$  mA, and  $i_3 = 50$  mA, find  $i_4$ ,  $i_5$ , and  $i_6$ .

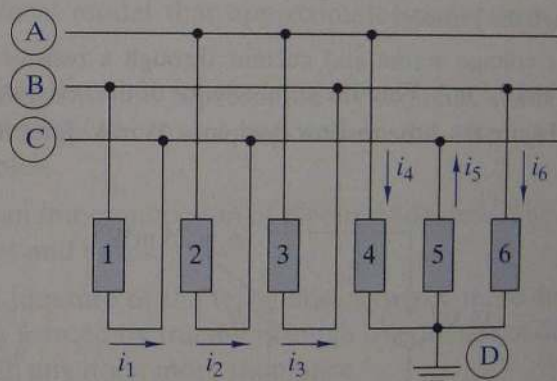


FIGURE P2-15

2-16 In Figure P2-16  $v_2 = 10$  V,  $v_3 = 10$  V, and  $v_4 = 10$  V. Find  $v_1$  and  $v_5$ .

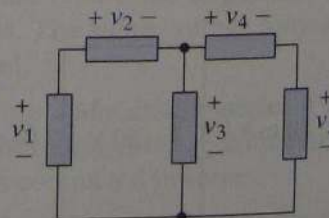


FIGURE P2-16



2-17 In Figure P2-17  $i_2 = 2$  A,  $i_3 = -5$  A, and  $i_4 = 4$  A. Find  $i_1$  and  $i_5$ .

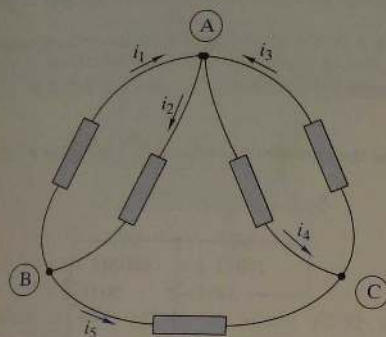


FIGURE P2-17

2-18 Use the passive sign convention to assign voltage variables consistent with the currents in Figure P2-17. Write three KVL connection equations using these voltage variables.

2-19 The KCL equations for a three-node circuit are:

$$\text{Node A: } -i_1 + i_2 - i_4 = 0$$

$$\text{Node B: } -i_2 - i_3 + i_5 = 0$$

$$\text{Node C: } i_1 + i_3 + i_4 - i_5 = 0$$

Draw the circuit diagram and indicate the reference directions for the element currents.

### OBJECTIVE 2-3 COMBINED CONSTRAINTS (SECT. 2-3)

Given a circuit consisting of independent sources and linear resistors, use the element constraints and connection constraints to find selected signal variables. See Examples 2-7, 2-8, 2-9, 2-10 and Exercise 2-6

2-20 Find  $v_x$  and  $i_x$  in Figure P2-20.

2-21 Find  $v_x$  and  $i_x$  in Figure P2-21.

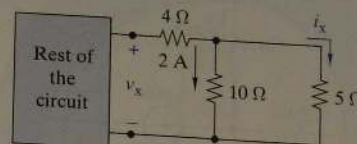


FIGURE P2-21

2-22 In Figure P2-22:

- Use the passive sign convention to assign a voltage and current variable to every element.
- Use KVL to find the voltage across each resistor.
- Use Ohm's law to find the current through each resistor.
- Use KCL to find the current through each voltage source.

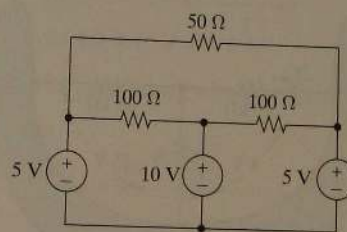
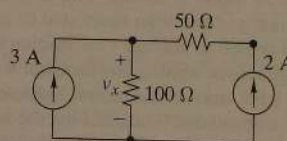


FIGURE P2-22

2-23 Find  $v_x$  in Figure P2-23.



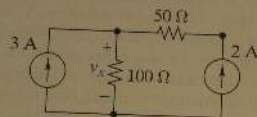


FIGURE P2-23

- 2-24 Figure P2-24 shows a subcircuit connected to the rest of the circuit at four points.  
 (a) Use element and connection constraints to find  $v_x$  and  $i_x$ .  
 (b) Show that the sum of the currents into the rest of the circuit is zero.

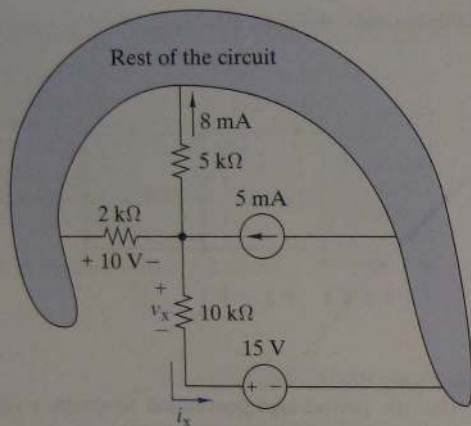
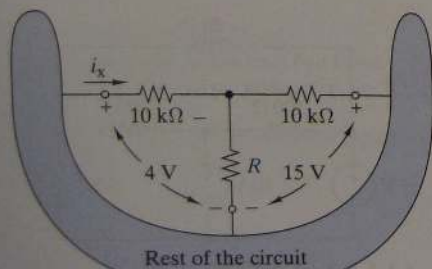


FIGURE P2-24

- 2-25 In Figure P2-25  $i_x = -0.5$  mA. Find the value of  $R$ .



### OBJECTIVE 2-4 EQUIVALENT CIRCUITS (SECT. 2-4)

- (a) Given a circuit consisting of linear resistors, find the equivalent resistance between a specified pair of terminals.  
 (b) Given a circuit consisting of a source-resistor combination, find an equivalent source-resistor circuit.  
 See Example 2-11, 2-12 and Exercises 2-7, 2-8, 2-9

- 2-27 Find the equivalent resistance  $R_{EQ}$  in Figure P2-27.

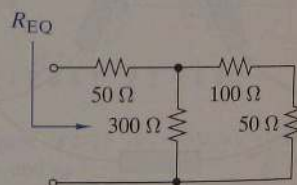


FIGURE P2-27

- 2-28 Find the equivalent resistance  $R_{EQ}$  in Figure P2-28.

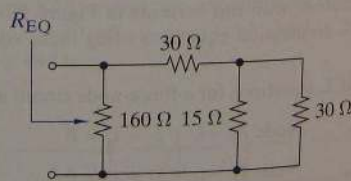


FIGURE P2-28

- 2-29 Find  $R_{EQ}$  in Figure P2-29 when the switch is open. Repeat when the switch is closed.

### PROBLEMS

- 2-31 In Figure P2-31 find the equivalent resistance between terminals A-B, A-C, A-D, B-C

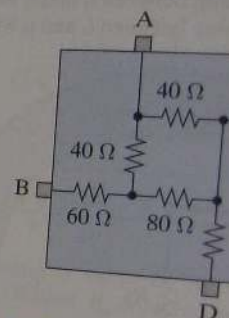


FIGURE P2-31

- 2-32 Select a value of  $R_L$  in Figure P2-32 such that  $R_{EQ} = 25$  kΩ.

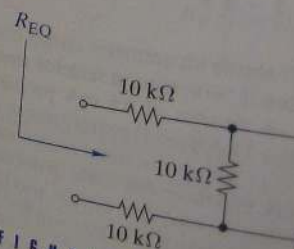


FIGURE P2-32

- 2-33 Repeat Problem 2-32 for  $R_{EQ} = 10$  kΩ.  
 2-34 Find the equivalent resistance between terminals A and B in Figure P2-34.